

Marking Rubric for Applied #2 Assessment (Week 10)

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APPLIED 2 Marking Rubric

		HD	D	C	P	N
Content (40%)						
Task 1 Image Rotation with GPU (20%)	1a) (6%)	Explanations on how image data can be transferred between the memory of a machine and the memory of a GPU are correct, without any omissions. Clear and readable diagrams are used to present how data is flowing between memory of a machine and GPU memory. All diagrams, supporting figures, and explanations are correct. No noticeable errors or component omissions can be found. All the explanations, diagrams, graphs, and arguments are clear, precise, and concise without any unnecessary details. Extensive use of parallel computing terminologies is required at this level.	Explanations on how image data can be transferred between the memory of a machine and the memory of a GPU are correct, without any omissions. Clear and readable diagrams are used to present how data is flowing between memory of a machine and GPU memory. All diagrams/supporting figures are correct and without any component missing or omissions.	Explanations on how image data can be transferred between the memory of a machine and the memory of a GPU are correct, without any major omissions. Clear and readable diagrams are used to present how data is flowing between memory of a machine and GPU memory. All diagrams/supporting figures are correct and without any major component missing or omissions.	Explanations on how image data can be transferred between the memory of a machine and the memory of a GPU are largely correct, despite some omissions/minor errors.	Explanations are not related to image data (or) Explanations are not related to GPU or memory (or) Explanations contains major errors or completely wrong (or) No attempt to the question

	1b) (7%)	Explanations on how the CUDA programming model operates are correct, without any omissions/errors. Explanations on the expected speed-up by modern GPU architectures are correct, without any errors. Clear and readable diagrams and sample codes are used to present how the CUDA programming model operates. Multiple sample codes are being crafted and used to illustrate different features of the CUDA programming models extensively. How different types of features would affect the speed-up are also clearly explained. All diagrams, supporting figures, and explanations are correct. No noticeable errors or component omissions can be found. All the explanations, diagrams, graphs, and arguments are clear, precise, and concise without any unnecessary details. Extensive use of parallel computing terminologies is required at this level.	Explanations on how the CUDA programming model operates are correct, without any omissions/errors. Explanations on the expected speed-up by modern GPU architectures are correct, without any errors. Clear and readable diagrams and sample codes are used to present how the CUDA programming model operates. All diagrams/supporting figures are correct and without any component missing or omissions.	Explanations on how the CUDA programming model operates are correct, without any major omissions/errors. Explanations on the expected speed-up by modern GPU architectures are correct, without any major errors. Clear and readable diagrams and sample codes are used to present how the CUDA programming model operates. All diagrams/supporting figures are correct and without any major component missing or omissions.	Explanations on how the CUDA programming model operates are correct, despite some omissions/minor errors.	Explanations are not related to the CUDA programming model (or) Explanations are not related to GPU (or) Explanations contains major errors or completely wrong (or) No attempt to the question
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	1c) (7%)	Explanations on how GPUDirect Storage would affect image rotation task and whether it would/would not benefit the task are correct. Research materials on GPUDirect Storage have been presented without any errors/omissions. Clear and readable diagrams are used to present how GPUDirect Storage operates. All diagrams, supporting figures, and explanations are correct. No noticeable errors or component omissions can be found. All materials are properly referenced and cited (in any major academic reference format). All the explanations, diagrams, graphs, and arguments are clear, precise, and concise without any unnecessary details. Extensive use of parallel computing terminologies is required at this level.	Explanations on how GPUDirect Storage would affect image rotation task and whether it would/would not benefit the task are correct. Research materials on GPUDirect Storage have been presented without any errors/omissions. Clear and readable diagrams are used to present how GPUDirect Storage operates. All diagrams/supporting figures are correct and without any component missing or omissions. All materials are properly referenced and cited (in any major academic reference format).	Explanations on how GPUDirect Storage would affect image rotation task and whether it would/would not benefit the task are correct. Research materials on GPUDirect Storage have been presented without any major errors/omissions. Clear and readable diagrams are used to present how GPUDirect Storage operates. All diagrams/supporting figures are correct and without any major component missing or omissions.	Explanations on how GPUDirect Storage would affect image rotation task are largely correct, despite some omissions/minor errors.	Explanations are not related to GPUDirect Storage (or) Explanations are not related to GPU (or) Explanations contains major errors or completely wrong (or) No attempt to the question
Task 2 Ethical Implications of Scaling HPC for AI (20%)	Content Correctness, Completeness, and Accuracy (20%)	Presentation covers all the 4 required topics listed in the specification. All the topics covered are sound and logically correct, and without significant/major materials missing/lacking. Extensive research has	Presentation covers all the 4 required topics listed in the specification. All the topics covered are sound and logically correct, and without significant/major materials missing/lacking. Extensive research has	Presentation covers all the 4 required topics listed in the specification. All the topics covered are sound and logically correct, and without significant/major materials missing/lacking. Materials are sourced from	Presentation covers all the 4 required topics listed in the specification. All the topics covered are largely sound, despite some minor errors. Materials are sourced from credible and traceable	Presentation is not related to HPC and its scaling issues (or) Presentation is not related to ethical/environmental/social issues (or) More than 70% of the

		been performed to introduce up-to-date materials. Extensive credible research articles are being used (for each topic) to provide insightful information and allow the presentation to draw informative comparisons. Materials are sourced from credible and traceable resources, and all the materials are properly cited/referenced (in any major academic reference format). Diagrams, figures, and pictures are extensively being used to improve presentation quality. All diagrams/figures/pictures are correct. All the explanations, diagrams, graphs, and arguments are clear, precise, and concise without any unnecessary details. Extensive use of parallel computing terminologies is required at this level.	been performed to introduce up-to-date materials. Materials are sourced from credible and traceable resources, and all the materials are properly cited/referenced (in any major academic reference format). Diagrams, figures, and pictures are extensively being used to improve presentation quality. All diagrams/figures/pictures are correct.	credible and traceable resources, and all the materials are properly cited/referenced (in any major academic reference format).	resources.	presentation is off-topic, out-of-scope, or incorrect. (or) Presentation is based on materials that are not from credible sources, or from unknown/non-traceable sources.
Presentation Quality and Q&A (60%)						
Presentation Quality (Clarity and Organisation) (20%)	The presentation is between 6 to 7 minutes. The presentation is <u>exceptionally</u> clear, well-organised, and easy to follow. The sections summarise the	The presentation is between 6 to 7 minutes. The presentation is clear and organised with minor areas for improvement. The sections summarise the presentation, and discuss	The presentation is generally understandable but has major rooms for improvements.	The presentation is somewhat understandable but lacks either organisation or content flow.	The presentation is largely unclear, poorly organised, and hard to follow.	

	presentation well, and discuss major issues, limitations, and future work. The presentation is concise, precise, and accurate (i.e., materials have been digested, analyzed, and chosen carefully without unnecessary information or details). Students are required to extensively use parallel computing terminologies (instead of general computing terms) at this level.	major issues, limitations, and future work.			
Q&A (40%)	Capable of understanding and answering all the questions without external tools. All the answers are correct without any flaws. The answers are concise, precise, and accurate (i.e., fully answer the question without unnecessary information or details). Students are required to answer by using parallel computing terminologies (instead of general computing terms) at this level.	Capable of understanding all the questions and answering all the questions without external tools. The answers are all correct without any flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some minor flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some major flaws.	Fails to understand the questions (or) Relies on external tools (e.g., AI, software tools, or online tools) to understand the questions or answer the questions. (or) Answers are completely wrong with major flaws/issues.

Special Consideration & Extensions

- From the special consideration procedure, short extensions are not available for group assessments, in-class assessments (including in-class tests/quizzes and presentations), practical assessments (including laboratories), mid-semester tests, or scheduled final assessments. For these assessment types, students must instead apply for special consideration.
- Unless a special consideration extension is approved, the student group/team will generally receive a 0 mark if they fail to attend the in-class session, fail to demonstrate the work, or fail to submit the required files to Moodle.
- For students with special considerations approved by the special consideration team, an alternative form of assessment must be submitted no later than the newly approved deadline. Students are reminded to read and follow the special consideration and alternate assessment sections in the assessment specification.
- If the extended due date falls after Week 14 and you plan to submit on/after Week 14, students are advised to contact the Chief Examiner and/or the Unit Coordinator, as grades may need to be withheld to allow extra time for marking.
- Students who have received special considerations but failed to submit the required files to Moodle by the newly approved deadline for their alternative assessment will be treated as having made no submission (or as having submitted late, if the file eventually arrives after the newly approved due date). For alternative assessments, late penalties will still apply for late submissions.